Group Project

Developing a MATSim Scenario

1 Overview

Students are asked to complete a group project in which they will use MATSim to develop several scenarios. There will be **5-6 students** in each group.

In this project, students will be given two base MATSim scenarios. Each group can choose **one of them** as a base to develop their own MATSim scenarios. The input data for these two base scenarios (i.e., two zip files) are available on the Blackboard in the following directory: *Blackboard → Content → Group Project*

Scenario 1: equil

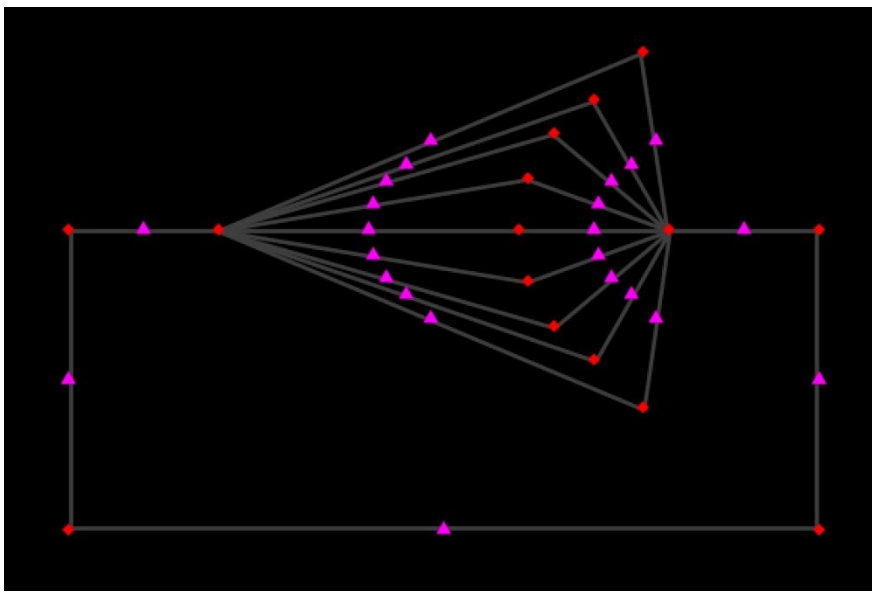


Figure 1 Network and Facilities in Scenario 1

Scenario 2: chessboard

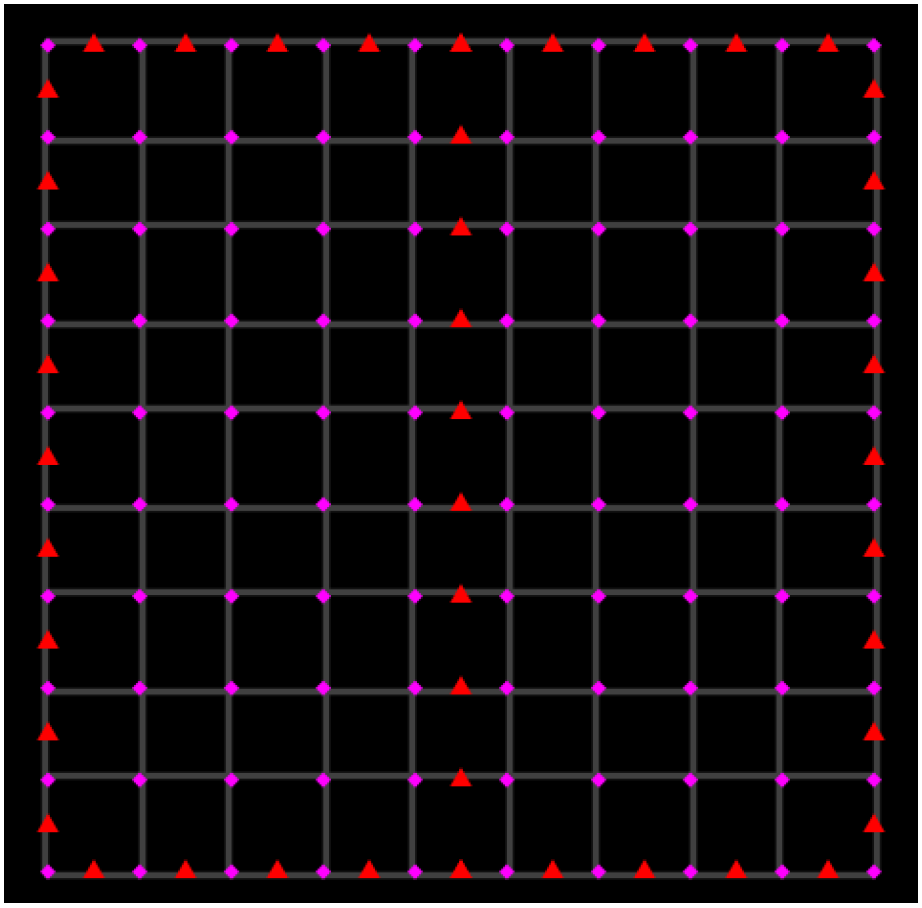


Figure 2 Network and Facilities in Scenario 2

2 Essential Tasks of the Group Project

This group project includes the following essential tasks:

- **Choosing one of the two given scenarios and then developing your own MATSim scenarios by modifying the following elements:**

  (1) Modifying input data (e.g., network, facility and daily plan)

  (2) Modifying configuration (e.g., config.xml)

  Note: you should explain why you make these modifications.

  Example: Increasing the capacity of a road link in the road network in order to test how changing the capacity of a road link would influence the travel behavior of

citizens.

- **Running those MATSim scenarios that you develop, and analyzing the simulation results of each MATSim scenario**

  Your analyses should include:

  (1) Trip Information (e.g., number of trips, departure time)

  (2) Score of Daily Plan (e.g., average score)

  (3) Link States

  (4) *You may wish to include other interesting results*

  Note: You may need to compare the results of your MATSim scenarios against those from the base scenario that you choose, in order to understand how your modification could influence the simulation outcomes.

- **Writing a report** introducing your group project. This should include:

  (1) Title

  (2) Abstract

  (3) Background

  (4) Introduction to MATSim (e.g., framework, mechanisms)

  (5) Introducing the base MATSim scenario that you choose, and also describing your own MATSim scenarios **(e.g., how and why** do you modify the key elements, such as network and daily plan, in your scenarios).

  (6) Results of your MATSim scenarios (Note: you may need to compare the results of your MATSim scenarios against those from the base scenario that you choose, in order to understand how your modifications could influence the simulation outcomes).

  (7) Conclusions (e.g., key findings, limitations and future work)

  (8) References (NO specific requirement on reference format)

- **Presenting** your group project in our last two labs and lectures of this subject. Each group will have **15 minutes** to present their projects. We will have Q & A session

after each presentation. Scheduled time slots for the presentation are as follows:

- ✓ 08:30-10:20, 18-Nov-25 (Venue: FJ302)
- ✓ 12:30-14:20, 18-Nov-25 (Venue: ZN604)
- ✓ 08:30-10:20, 25-Nov-25 (Venue: FJ302)
- ✓ 12:30-14:20, 25-Nov-25 (Venue: ZN604)

3 Specific Requirements for Group Project

- This is a group project (50% of the overall grade). The students can form groups by themselves. For each group, it should have **one Group Leader** who coordinate between **four or five Members** (In total, 5-6 students in each group). *Group Leader* can download the form titled "*Group Members-Template.xlsx*" from "LSGI3213-ITS-Group Members" in the "Assessments". Please fill in the form and submit it through "LSGI3213-ITS-Group Members" in the "Assessments" by **3 Oct. 2025.** For those students who cannot find any group, they will be assigned to a group in due course.
- **Each group needs to submit a report, presentation slides, and team members' contribution (Optional):**

  (1) Report Requirement (see above for the report structure suggested): About 15 pages for each report, not more than 30 pages (Font Size: 12; Single Space)**.** *References should NOT be counted towards page requirement.* **File Name:** Group ID_NameOfGroupLeader_Report.pdf (e.g., Group 03_San Zhang_Report.pdf). *Note that a Group ID will be assigned to each group later*.

  (2) Requirements of Presentation Sides: Each group needs to prepare **slides** for their projects (for example, using Microsoft PowerPoint). **File Name:** Group ID_NameOfGroupLeader_Slides.pdf (e.g., Group 03_San Zhang_Slides.pdf)

  (3) Presentation Requirements: Each group needs to prepare a **15-Minute** presentation for their projects using the slides above. You will be asked to present your group project in our last two labs and lectures of this subject. Each group

could have **one or more** members to present their projects.

(4) Team Members' Contribution (Optional): This form is **optional**. You may wish to use this form when you think the contribution of any team member is much lower. The information in this form should be confirmed by ALL team members before being submitted. You can find a **template** (i.e., Team Members' Contribution_Optional.docx) on the Blackboard in the following directory: *Blackboard → Content → Group Project*. **File Name:** Group ID_NameOfGroupLeader_Team Members' Contribution.pdf (e.g., Group 03_San Zhang_Team Members' Contribution.pdf)

- **Deadline for Submitting Report, Slides and Team Members' Contribution (Optional):** *Group Leader* should submit their **report, slides and Team Members' Contribution (Optional)** through "LSGI3213-ITS-Group Project" in the "Assessments" of the Blackboard **by 14 Nov. 2025 (before 24:00)**. Note that this submission link will be available after **24 October 2025**.

<u>Important Notes:</u>

Generative AI (e.g., ChatGPT) can only serve as a tool for assisting initial idea development and proofreading for project presentation and report, and **any involvement of generative AI tools must be clearly acknowledged and referenced in your presentation/report**.